RL Project 1

Pranav Shukla

1 System Dynamics

The continuous-time dynamics are given by:

$$\begin{aligned}
\dot{p}_x &= V_x \\
m\dot{V}_x &= -(U_1 + U_2)\sin\theta \\
\dot{p}_y &= V_y \\
m\dot{V}_y &= (U_1 + U_2)\cos\theta - mg \\
\dot{\theta} &= \omega \\
I\dot{\omega} &= r(U_1 - U_2)
\end{aligned}$$

1.1 Discretisation using Euler's

We have,

$$\begin{aligned}
p_{x,n+1} &= p_{x,n} + V_{x,n}\Delta t \\
V_{x,n+1} &= V_{x,n} - \frac{(U_{1,n} + U_{2,n})\sin\theta_n \Delta t}{m} \\
p_{y,n+1} &= p_{y,n} + V_{y,n}\Delta t \\
V_{y,n+1} &= V_{y,n} + \frac{((U_{1,n} + U_{2,n})\cos\theta_n - mg)\Delta t}{m} \\
\theta_{n+1} &= \theta_n + \omega_n \Delta t \\
\omega_{n+1} &= \omega_n + \frac{r(U_{1,n} - U_{2,n})\Delta t}{I}
\end{aligned}$$

Let the state vector be $x = [p_x \ V_x \ p_y \ V_y \ \theta \ \omega]^T$. The matrix operation above is represented by a function f^n:

$$x_{n+1} = f(x_n, U_n) \quad \forall n \in [0, N-1]$$

$$x_0 = x_{init}$$

2 Linearization

We have 2 non-linear terms, let us linearise them about the current trajectory (x^*, U^*).

Let $T_1 = \frac{(U_{1,n}+U_{2,n})\sin\theta_n \Delta t}{m}$.

$$\begin{aligned} mT_1^{lin} = &-(U_{1,n}^* + U_{2,n}^*)\sin\theta_n^*\Delta t - (U_{1,n}^* + U_{2,n}^*)\cos\theta_n^*(\theta_n - \theta_n^*)\Delta t - \sin\theta_n^*\Delta t(U_{1,n} - U_{1,n}^*) \\ &- \sin\theta_n^*\Delta t(U_{2,n} - U_{2,n}^*) \end{aligned}$$

Let $U_{1,n} - U_{1,n}^* = \Delta U_{1,n}$
Let $U_{2,n} - U_{2,n}^* = \Delta U_{2,n}$
Let $\theta_n - \theta_n^* = \Delta\theta_n$
Let $-(U_{1,n}^* + U_{2,n}^*)\sin\theta_n^*\Delta t = mT_1^*$

$$mT_1^{lin} = mT_1^* - \sin\theta_n^*\Delta t\Delta U_{1,n} - (U_{1,n}^* + U_{2,n}^*)\cos\theta_n^*\Delta t\Delta\theta_n - \sin\theta_n^*\Delta t\Delta U_{2,n}$$

Similarly, Let $T_2 = \frac{((U_{1,n}+U_{2,n})\cos\theta_n - mg)\Delta t}{m}$.

$$mT_2^{lin} = mT_2^* + \cos\theta_n^*\Delta t(\Delta U_{1,n} + \Delta U_{2,n}) - (U_{1,n}^* + U_{2,n}^*)\sin\theta_n^*(\Delta t)\Delta\theta_n$$

2.1 Linearized State Dynamics

We replace all states using the following convention: $Y_n = Y_n^* + \Delta Y_n$. So we have, (using the linearised terms) (Note: * denotes current optmial trajectory):

$$\begin{aligned} \Delta p_{x,n+1} &= \Delta p_{x,n} + \Delta V_{x,n}\Delta t + p_{x,n}^* + V_{x,n}^*\Delta t - p_{x,n+1}^* \\ \Delta V_{x,n+1} &= \Delta V_{x,n} + T_1^* - \frac{\sin\theta_n^*\Delta t}{m}(\Delta U_{1,n} + \Delta U_{2,n}) \\ &\quad - \frac{(U_{1,n}^* + U_{2,n}^*)\cos\theta_n^*\Delta t}{m}\Delta\theta_n + V_{x,n}^* - V_{x,n+1}^* \\ \Delta p_{y,n+1} &= \Delta p_{y,n} + \Delta V_{y,n}\Delta t + p_{y,n}^* + V_{y,n}^*\Delta t - p_{y,n+1}^* \\ \Delta V_{y,n+1} &= \Delta V_{y,n} + T_2^* + \frac{\cos\theta_n^*\Delta t}{m}(\Delta U_{1,n} + \Delta U_{2,n}) \\ &\quad - \frac{(U_{1,n}^* + U_{2,n}^*)\sin\theta_n^*\Delta t}{m}\Delta\theta_n + V_{y,n}^* - V_{y,n+1}^* \\ \Delta\theta_{n+1} &= \Delta\theta_n + \Delta\omega_n\Delta t + \theta_n^* + \omega_n^*\Delta t - \theta_{n+1}^* \\ \Delta\omega_{n+1} &= \Delta\omega_n + \frac{r(\Delta U_{1,n} - \Delta U_{2,n})\Delta t}{I} + \omega_n^* + \frac{r(U_{1,n}^* - U_{2,n}^*)\Delta t}{I} - \omega_{n+1}^* \end{aligned}$$

(Note: The terms $f(x_n^*, U_n^*) - x_{n+1}^*$ are included above.)

3 QP Equality constraints(Linearized)

This linearised solution formulations the equality constraints for out QP inside the SQP. We can rewrite the equations from the above solution in matrix form as follows:

$$\Delta x_{n+1} = f(x^*, U^*) - x_{n+1}^* + A\Delta x_n + B\Delta U_n$$

$$A = \begin{bmatrix} 1 & \Delta t & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{-\Delta t}{m}(U_{1,n}^* + U_{2,n}^*)\cos\theta_n^* & 0 \\ 0 & 0 & 1 & \Delta t & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{-\Delta t}{m}(U_{1,n}^* + U_{2,n}^*)\sin\theta_n^* & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ \frac{-\Delta t \sin\theta_n^*}{m} & \frac{-\Delta t \sin\theta_n^*}{m} \\ 0 & 0 \\ \frac{\Delta t \cos\theta_n^*}{m} & \frac{\Delta t \cos\theta_n^*}{m} \\ 0 & 0 \\ \frac{r\Delta t}{I} & \frac{-r\Delta t}{I} \end{bmatrix}$$

4 Inequality Constraints for QP

The constraints are of the form $h \leq 0$. Let $U_1 = U_1^* + \Delta U_1$, $U_2 = U_2^* + \Delta U_2$, and $p_y = p_y^* + \Delta p_y$. The constraints are:

- $U_1 \geq 0 \implies -U_1^* - \Delta U_1 \leq 0$
- $U_2 \geq 0 \implies -U_2^* - \Delta U_2 \leq 0$
- $U_1 \leq 10 \implies U_1^* + \Delta U_1 - 10 \leq 0$
- $U_2 \leq 10 \implies U_2^* + \Delta U_2 - 10 \leq 0$
- $p_y \geq 0 \implies -p_y^* - \Delta p_y \leq 0$ (Bonus Constraint)

$$h(x_k, U_k) = \begin{bmatrix} -U_1 \\ -U_2 \\ U_1 - 10 \\ U_2 - 10 \\ -p_y \end{bmatrix}$$

The QP for the SQP will have the constraints:

$$\nabla h(x_k^*, U_k^*)^T p + h(x_k^*, U_k^*) \leq 0$$

where p is the optimization variable vector for this timestep, $p = \begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix}$.

So, for the QP vector p, we have:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} p \leq \begin{bmatrix} U_1^* \\ U_2^* \\ 10 - U_1^* \\ 10 - U_2^* \\ p_y^* \end{bmatrix}$$

5 SQP Cost Function

We take θ as unwrapped (No limit) and propose the following cost function:

$$C = \sum_{n=0}^{T-1} (\frac{1}{2}(x - x_n)^T K (x_n - x_d) + \frac{1}{2} U_n^T G U_n) + \frac{1}{2} x_T^T M x_T$$

We take the the target X_d as

$$x_d = [0\ 0\ 0\ 0\ 2\pi\ 0]^T$$

- K penalises the states during the trajectory , primarty the angle theta $K = diag(4, 3, 5, 3, 6, 1)$
- M penalises V_x, V_y, θ, ω, whith smaller weights for the others, to penalised the terminal constraints, focusing on minimum velocities to retain stability. $M = diag(4, 20, 5, 20, 6, 1)$
- G penalises U_1, U_2 , the control inputs. $(G = diag(3, 3))$

6 Sequential Quadratic Programming (SQP) Formulation

The QP cost inside the sqp is problem is to:

$$\min_{\Delta x, \Delta U} \sum_{k=0}^{T-1} \left(\frac{1}{2} \begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix}^T \begin{bmatrix} Q_k & S_k \\ S_k^T & R_k \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix} + \begin{bmatrix} q_k \\ r_k \end{bmatrix}^T \begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix} \right) + \frac{1}{2} \Delta x_T^T Q_T \Delta x_T + \Delta x_T^T q_T$$

Here, we ignore the hessian of the dynamaics and have:

$$\begin{aligned} q_k &= \nabla_x l = K(x_k^* - x_d) \\ q_T &= \nabla_x l_T = M x_T^* \\ Q_T &= \nabla_{xx} l_T = M \\ Q_k &= \nabla_{xx} l_k = K \\ S_k &= \nabla_{xu} l_k = 0 \\ R_k &= \nabla_{uu} l_k = G \\ r_k &= \nabla_u l_k = G U_k^* \end{aligned}$$

So $\rightarrow$

$$\min_{\Delta x, \Delta U} \sum_{k=0}^{N-1} \left(\begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix}^T \begin{bmatrix} K & 0 \\ 0 & G \end{bmatrix} \begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix} + \begin{bmatrix} K(x_k^* - x_d) \\ G U_k^* \end{bmatrix}^T \begin{bmatrix} \Delta x_k \\ \Delta U_k \end{bmatrix} \right) + \frac{1}{2} \Delta x_T^T M \Delta x_T + \Delta x_T^T M x_T^*$$

7 Solving the SQP

- We convert the equality and inequality constraint to a larger matrix form that encompasses trajectory states and controls into a vector p.
- We then convert the Cost summation also as a quadratic and linear Matrix multiplication operation over all the points in the trajectory.
- We then feed the cost, linear equality, inequality matrices to the qp solver cvxopt, and get a solution for p.
- We use filter line search to find the step size alpha, and then update the current trajectory guess X, and repeat until convergence.
- One small optimisation was the target thrust values were set to enable hover , each set to $m * g/2$

8 MPC Setup

- for the MPC, we use the same cost function and X_d target as the SQP.
- Inside the controller function, we start to initialize the trajectory X , with the first state given by the controller input x, and the rest as zeros, and the size of X is the MPC horizon, taken as 35.
- We now run the SQP to optimize this trajectory. We found that since the MPC is iterative, we do not necessarily need to converge this SQP. We are running this for 5 iterations.
- After the SQP has updated the trajectory, we take the first pair of controls and output that to the simulator.
- We roll the current trajectory X over by 1 timesteps to be utilised as a warm start for the next iteration. In the next iteration, we change the first state to the state given into the controller
- We run this MPC for 250 timesteps (10 seconds)

9 Results of the SQP:

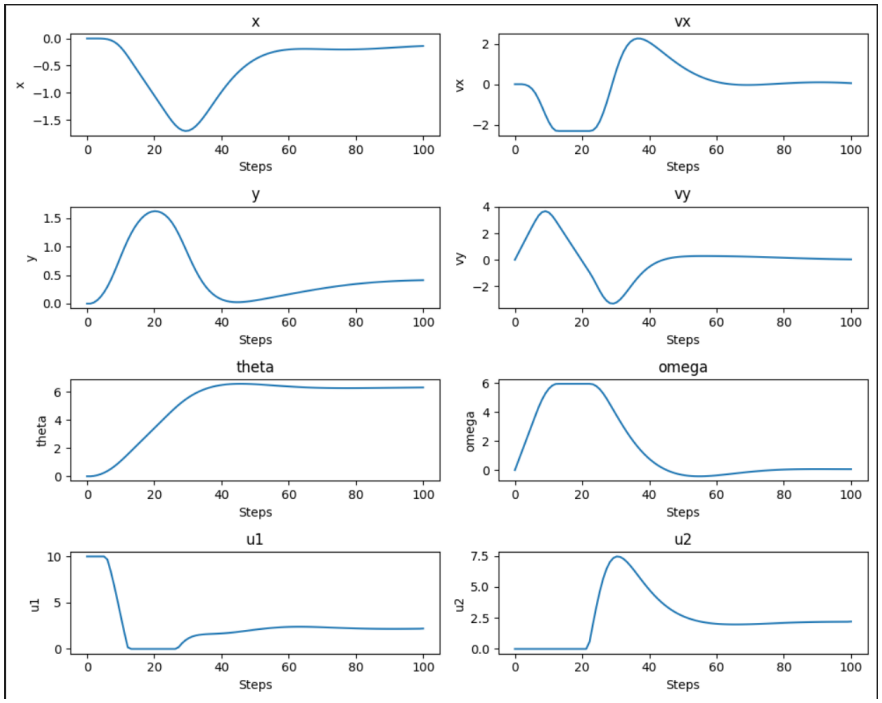

Figure 1: State and Control Input Results from the SQP

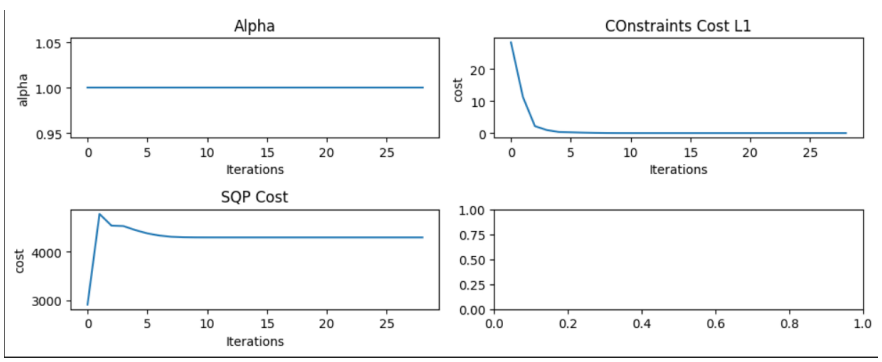

Figure 2: Step Size and Constraint and SQP Costs

10 Results of the MPC with disturbances:

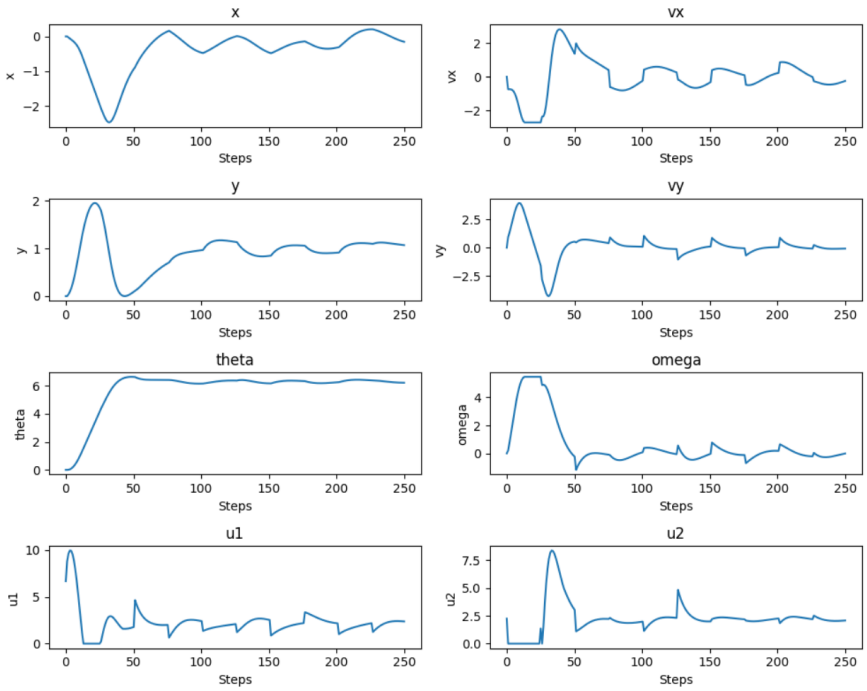

Figure 3: State and Control Input Results from the SQP